

Annotated Bibliography

Yi, S., Liu, Y., Sun, Z., Cong, T., He, X., Song, J., Xu, K., & Li, Q. (2024). Jailbreak Attacks and Defenses Against Large Language Models: A Survey. arXiv preprint arXiv:2407.04295. <https://arxiv.org/abs/2407.04295>

This comprehensive survey provides a detailed taxonomy of jailbreak attack and defense methods against Large Language Models (LLMs). The authors categorize attack methods into black-box and white-box attacks based on target model transparency, while classifying defense methods into prompt-level and model-level defenses. The paper presents a coherent diagram illustrating relationships between different attack and defense sub-classes, investigates current evaluation methods, and compares them from multiple perspectives. Submitted on July 5, 2024, and revised on August 30, 2024 (v2), this 1,338 KB survey aims to enhance understanding of the jailbreak domain and provide a foundation for developing more secure LMs despite jailbreak remaining a significant community concern.

Chen, G., Xia, Y., Jia, X., Li, Z., Torr, P., & Gu, J. (2025). LLM Jailbreak Detection for (Almost) Free! arXiv preprint arXiv:2509.14558. <https://arxiv.org/abs/2509.14558>

This paper introduces Free Jailbreak Detection (FJD), a method that detects jailbreak prompts with almost no additional computational costs during LLM inference. The authors present a key finding that the difference in output distributions between jailbreak and benign prompts can be employed for detection. FJD prepends an affirmative instruction to the input and scales logits by temperature to distinguish between jailbreak and benign prompts through the confidence of the first token. The method is enhanced through virtual instruction learning integration. Extensive experiments on aligned LLMs demonstrate effective jailbreak detection. Accepted to EMNLP 2025 (Findings), this paper was submitted on September 18, 2025, and revised on January 23, 2026 (v2).

Yi, J., Xie, Y., Zhu, B., Kiciman, E., Sun, G., Xie, X., & Wu, F. (2023). Benchmarking and Defending Against Indirect Prompt Injection Attacks on Large Language Models. arXiv preprint arXiv:2312.14197. <https://arxiv.org/abs/2312.14197>

This paper introduces the first benchmark for indirect prompt injection attacks, named BIPIA, to assess vulnerabilities in LLMs integrated with external content. The authors evaluate existing LLMs and find them universally vulnerable to malicious instructions embedded within external content that manipulate LLM outputs. Analysis identifies two key success factors: LLMs' inability to distinguish between informational context and actionable instructions, and lack of awareness in avoiding execution of instructions within external content. The paper proposes two novel defense mechanisms boundary awareness and explicit reminder for both black-box and white-box settings. Experiments demonstrate substantial mitigation in black-box defense and near-zero attack success rates in white-box defense while preserving output quality. Accepted by KDD 2025, submitted December 21, 2023, revised January 27, 2025 (v4).

Hung, K.-H., Ko, C.-Y., Rawat, A., Chung, I.-H., Hsu, W. H., & Chen, P.-Y. (2024). Attention Tracker: Detecting Prompt Injection Attacks in LLMs. arXiv preprint arXiv:2411.00348. <https://arxiv.org/abs/2411.00348>

This paper investigates prompt injection attack mechanisms by analyzing attention patterns within LLMs, introducing the concept of the "distraction effect" where specific attention heads (important heads) shift focus from original instructions to injected instructions. The authors propose Attention Tracker, a training-free detection method that tracks attention patterns on instructions to detect prompt injection attacks without additional LLM inference. The method generalizes effectively across diverse models, datasets, and attack

types, showing AUROC improvement up to 10.0% over existing methods and performing well even on small LLMs. Extensive evaluations demonstrate robustness and provide insights for safeguarding LLM-integrated systems. Submitted November 1, 2024, revised April 23, 2025 (v2).

Shaheer, S., Islam, G. M. R., Hamid, M. R., Khan, M. A. F., Faruk, M. O., & Nur, Y. (2025). Detecting Prompt Injection Attacks Against Application Using Classifiers. arXiv preprint arXiv:2512.12583. <https://arxiv.org/abs/2512.12583>

This undergraduate research project curates and augments a prompt injection dataset based on the HackAPrompt Playground Submissions corpus to detect malicious prompts in LLM-integrated web applications. The authors train several classifiers including LSTM, feed-forward neural networks, Random Forest, and Naive Bayes to detect prompt injection attacks that compromise security and stability of critical systems from infrastructure to large web applications. The proposed approach improves prompt injection detection and mitigation, helping protect targeted applications and systems. This 9-page paper was submitted on December 14, 2025 (v1).